

3D Scanning Electron Microscope and X-ray fluorescence on volcanic ash micro-surfaces from Iceland

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Abstract

Portable X-ray fluorescence (pXRF) and benchtop scanning electron microscope equipped with energy dispersive system (SEM-EDS) are two analytical procedures that can be used to examine geological rocks. Samples were collected from the Eyjafjallajökull volcano that recently erupted in 2010, Grimsvötn volcano, and Vatnafjöll volcano in Iceland and analyzed under a scanning electron microscope for geochemical and physical properties. SEM images were imported into MountainMaps and TopoMaps 3D software to analyze linear features and render a binary outline to be printed. Direction and the frequency of those linear features were statistically examined. First, second and third direction peaks were obtained from a rose diagram that may imply the directional lava flow of the volcanic rock cooling and deposition procedure during the volcanic eruption of the Eyjafjallajökull volcano in Iceland. pXRF tests indicate compositional similarities between volcanic ash samples as well as elemental homogeneity with limited exception.

Figure 1: Aerial image obtained from Google Earth of the study area and the sampling locations on Iceland. Sampling locations are shown with yellow pins.



Introduction - Methods

Physical properties of volcanic rocks may reveal the way that lava was deposited and behaved at the time of the volcanic eruption of 2010 at Eyjafjallajökull volcano. Using micro-scale analytical techniques such as SEM, EDS, and pXRF the micro-surfaces of the volcanic rocks may be investigated.

SEM

A scanning electron microscope (SEM) is a type of electron microscope that produces images of a sample by scanning the surface with a focused beam of electrons. The electrons interact with atoms in the sample, producing various signals [1,2] that contain information about the surface topography and composition of the sample.

EDS

Energy Dispersive X-Ray Spectroscopy (EDS or EDX) is a chemical microanalysis technique used in conjunction with scanning electron microscopy (SEM). The EDS technique detects x-rays emitted from the sample during bombardment by an electron beam to characterize the elemental composition of the analyzed volume [3].

P-XRF

X-ray Fluorescence is a technique for chemical compositional measurement. X-rays of a known energy are directed towards a target or sample, causing the atoms within the material to emit "fluorescent" X-rays at energies characteristic of its elemental composition [4].

Results

Units: ppm	Nb	Zr	Sr	Rb	Hf	Fe	Mn	Ti	Al	Si	Cl	S	Mg	MgO
E1	84.36	618.65	387.64	33.17	169.95	98094.66	2071.61	13156.55	6444.05	63379.74	547.41	<LOD	4364.65	7245.32
G1	50.56	390.32	295.08	18.62	202.96	104357.97	1886.44	14445.38	5329.63	44044.57	500.53	<LOD	<LOD	1.66
V1	42.94	354.87	326.67	17.01	104.95	107142.92	2082.82	20769.77	52998.84	170466.67	2812.07	1027.14	18343.92	30450.91

Table 1: Data retrieved from X-ray fluorescence indicate similarities and elemental differences between 3 different volcanic ash samples



Figure 2: Achek (Momo) Moread and Dr. Marsellos prepare the samples for analysis under the scanning electron microscope (SEM).

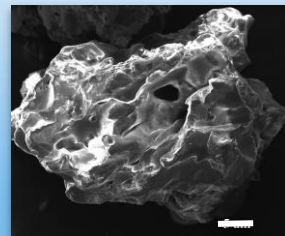


Figure 3: Scanning electron microscope image of a volcanic rock sample from the Eyjafjallajökull volcano in Iceland.



Figure 4: (left) The volcanic sample collected from the Eyjafjallajökull volcano is mounted on the micro-stages for analysis under the scanning electron microscope (SEM).

Table 2: MountainMaps and TopoMaps 3D software has provided a rose diagram. Rose diagram allows data visualization of direction and frequency or intensity.

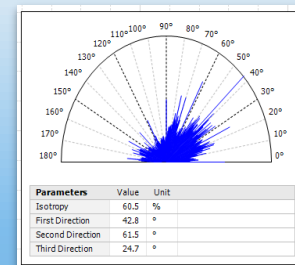
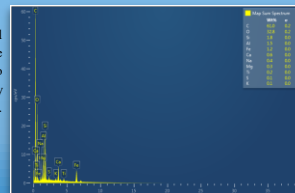


Figure 5: (right) Geochemical analysis of a volcanic ash sample from the Eyjafjallajökull volcano under the Energy Dispersive X-Ray spectroscopy (EDS).



Discussion

- Direction analysis may potentially result to significant conclusions about the direction of a lava flow during the eruption episode.
- First and second direction results may provide interpretation on the cooling procedure and geochemical composition of a lava during deposition.
- Further research must be conducted in order to correlate compositional data with that of other volcanic ash samples

Conclusion

Microscopic analysis has shown that linear features on the micro-surface of the volcanic ash reveal a dominant orientation. This physical characteristic may imply the direction of the lava flow.

References

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